IPO Intelligence

The AI Research Terminal for IPO Investors

Project Category: FinTech / AI / RAG
Target Market: Indian IPO Investors
Core Technology: MongoDB Atlas Vector Search, Hybrid Search and LLMs

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1. Problem Statement

Evaluating an IPO requires analysing information scattered across lengthy DRHPs, RHPs, financial statements, competitor reports, regulatory filings, news articles and market data.

Investors often struggle to:

- Understand a company's actual financial health.
- Identify hidden risks in lengthy prospectuses.
- Evaluate IPO valuation against competitors.
- Distinguish genuine business strength from market hype.
- Connect financial fundamentals with subscription trends and GMP.
- Make informed decisions using reliable, verifiable information.

IPO Intelligence aims to solve this information fragmentation problem through an AI-powered, evidence-backed IPO research platform.

2. Proposed Solution

IPO Intelligence is an AI-powered research platform that collects, processes and analyses IPO-related information from multiple sources.

Users can select an IPO, explore its fundamentals, investigate risks, compare valuations and ask complex questions in natural language.

The platform uses Retrieval-Augmented Generation (RAG) with MongoDB Vector Search to retrieve relevant information from multiple documents and generate answers with direct citations to original sources.

Its objective is not merely to provide an APPLY / MAYBE / SKIP recommendation, but to explain the reasoning and evidence behind it.

3. Core Product Modules

Module| Key functionality
IPO Dashboard| Upcoming IPOs, issue details, GMP and subscription snapshots
Company Fundamentals| Revenue, EBITDA, PAT, margins, debt and cash flows
Valuation & Peer Comparison| P/E, EV/EBITDA and competitor analysis
Risk Intelligence| Litigation, promoter concerns, customer concentration and financial red flags
Market Sentiment| GMP trends, QIB/NII/Retail subscription and demand analysis
AI Research Analyst| Natural-language questions and multi-document investigation
Report Generator| Comprehensive IPO research reports with citations

4. Data Sources

The initial version will focus on Indian IPOs.

Primary sources

- SEBI – DRHPs, RHPs, prospectuses and regulatory filings.
- NSE and BSE – Issue information and market disclosures.
- Company annual reports and financial statements.
- Official company announcements.

Secondary sources

- Screener – Financial data and competitor analysis.
- Moneycontrol – Financial news and market information.
- Chittorgarh – IPO information and subscription details.
- Other relevant industry and market reports.

Data collection strategy: Start with a curated collection of approximately three IPOs, their prospectuses, historical financial statements, competitor information and supporting articles. Support user uploads and selected public sources as well.

GMP and subscription information should be stored as timestamped observations because they change over time.

5. Key RAG Capabilities

The RAG pipeline will support:

1. Document ingestion: Upload and process PDFs, financial reports and CSV/XLSX files.
2. Intelligent chunking: Preserve document sections, tables, page numbers and headings.
3. Embedding generation: Convert document chunks into numerical vector representations.
4. MongoDB Vector Search: Retrieve semantically relevant passages.
5. Hybrid search: Combine semantic retrieval with keyword search for exact financial terms and identifiers.
6. Metadata filtering: Filter by company, document type, financial year and publication date.
7. Multi-source reasoning: Combine evidence from prospectuses, financial reports and competitor documents.
8. Citation generation: Link important findings to exact source passages and pages.
9. Evidence verification: Identify conflicting information and unsupported claims.
10. Abstention: Clearly acknowledge when available information is insufficient.

6. Financial Analysis & Recommendation Engine

The platform will evaluate IPOs across five dimensions:

Evaluation dimension	Weight
Financial fundamentals	30%
Valuation	20%
Business and financial risks	25%
Market sentiment	15%
Listing potential	10%

These are initial proposed weights and will be calibrated through testing.

The platform will provide separate research assessments for:

- Listing gains: Valuation, subscription, GMP and market sentiment.
- Long-term investment: Fundamentals, competitive advantage, cash flows and governance.
- Capital protection: Balance-sheet strength, valuation risks and downside exposure.

The final output can include an APPLY / MAYBE / SKIP assessment, supported by transparent scoring, risks and evidence.

Financial calculations must use structured data and deterministic formulas rather than relying on LLM-generated numerical estimates.

7. MongoDB Architecture

MongoDB Atlas will serve as the central database.

Collection	Purpose
"ipos"	IPO details and issue information

"sources"| Original documents, URLs and metadata
"chunks"| Extracted text, embeddings and source references
"financials"| Structured financial metrics
"market_data"| GMP, subscription and listing observations
"peers"| Competitor information
"reports"| Generated research reports
"evaluations"| Test queries and expected source IDs

Core retrieval workflow:

User Query → Query Understanding → Vector Search + Keyword Search → Metadata Filtering → Result Fusion and Reranking → LLM Reasoning → Evidence-Backed Answer with Citations.

Original source documents should remain accessible so users can verify the evidence themselves.

8. Illustrative Hackathon Demo

The demonstration will revolve around a hypothetical company, ABC Industries.

Demo 1 – Multi-source investigation

"Revenue is growing, but operating cash flow is deteriorating. Investigate the reasons, evaluate its debt position and compare its valuation with competitors."

The system retrieves relevant financial statements, prospectus sections, risk disclosures and competitor reports to produce an evidence-backed analysis.

Demo 2 – Conflicting information

"An article claims the company has strong cash generation, but its prospectus shows negative operating cash flow. Explain the discrepancy."

The system compares reporting periods, publication dates and supporting evidence.

Demo 3 – Unanswerable question

"Will this IPO definitely list at a 25% premium?"

The system explains that a guaranteed listing price cannot be established from available evidence and offers scenario-based analysis instead.

Citation demonstration: Click a citation to open the original document and view the exact supporting passage.

9. MVP Development Priorities

Phase 1 – Core RAG

- Document upload and extraction.
- Chunking and embeddings.
- MongoDB integration.
- Vector Search and hybrid retrieval.
- Multi-source question answering.
- Page-level citations.

Phase 2 – Financial Intelligence

- Structured financial data extraction.
- Financial ratio calculations.
- Competitor comparison.
- Risk identification.
- APPLY / MAYBE / SKIP assessment.

Phase 3 – Product Experience

- IPO dashboard.
- Interactive citation explorer.
- Research report generation.
- Retrieval evaluation dashboard.
- Improved UI and user experience.

10. Proposed Technology Stack

Layer| Technology
Frontend| Next.js + TypeScript
UI| Tailwind CSS + shadcn/ui
Backend| Python + FastAPI
Database| MongoDB Atlas
Semantic retrieval| MongoDB Vector Search
Keyword retrieval| MongoDB Atlas Search
Embeddings| Embedding model API
LLM| LLM API
PDF processing| PyMuPDF
Deployment| Vercel + Render or equivalent

11. Key Differentiators

- Goes beyond a conventional PDF chatbot.
- Combines unstructured documents and structured financial data.
- Supports semantic and hybrid retrieval.
- Connects information across multiple sources.
- Identifies financial anomalies and potential red flags.
- Provides verifiable, page-level citations.
- Distinguishes facts, calculations, interpretations and hypotheses.
- Handles contradictory and insufficient evidence.
- Provides separate assessments for short-term and long-term investment objectives.

Product Vision

IPO Intelligence – The AI Research Terminal for IPO Investors

Understand the business. Verify the numbers. Uncover the risks. Evaluate the opportunity.

The goal is to build a reliable, transparent and intelligent IPO research assistant that transforms lengthy financial documents and fragmented market information into actionable, evidence-backed insights.

Core principle: Every important conclusion should have supporting evidence, every financial calculation should be verifiable, and every uncertainty should be communicated clearly.